Alexander **Elzenaar**

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Research interests

- Groups related to geometry and topology in low dimensions. Kleinian groups and their deformation theory; other aspects of geometric group theory, conformal geometry, and 3-manifold theory. Applications of techniques from representation theory and algebraic geometry.
- Exemplar MSC codes: 20F65, 20H10, 22E40, 30F40, 30F60, 32G15, 51F15, 57K31, 57K32.

Tertiary education

School of Mathematics, Monash University — **Melbourne, Australia**
Doctoral student, geometry and topology — *2024–ongoing*
Advisor: Prof. Jessica Purcell.
Expected completion: 2027.

Dept. of Mathematics, The University of Auckland — **Auckland, NZ**
MSc with First Class Honours in mathematics — *2021–2022*
Advisors: Dist. Prof. Gaven Martin (NZ Inst. Adv. Study., Massey Uni.) and Dr. Jeroen Schillewaert.
Thesis: *Deformation spaces of Kleinian groups* (`https://aelzenaar.github.io/msc_thesis.pdf`)

Dept. of Mathematics, The University of Auckland — **Auckland, NZ**
BSc (Hons) with First Class Honours in mathematics — *2020*
Advisor: Dr. Jeroen Schillewaert.
Dissertation: *Toric varieties* (`https://aelzenaar.github.io/hons/dissertation.pdf`)

School of Cultures, Languages and Linguistics, The University of Auckland — **Auckland, NZ**
Certificate of Languages — *awarded 2020*
Russian and Ancient Egyptian

The University of Auckland — **Auckland, NZ**
BSc, major in mathematics — *2017–2019*

The University of Canterbury — **Christchurch, NZ**
STAR programme — *2016*
Distance learning first year university mathematics for secondary school students

Employment history (research)

Ministry of Business, Innovation, and Employment — **Wellington, NZ**
Research and data analyst, Evidence & Insights branch — *2023–2024*
Mathematical modelling and statistics to support ministerial offices and policy evaluators with particular focus in carbon modelling and econometric forecasting for the building and construction industry, including production and peer review of departmental research output.

Max-Planck-Inst. für Math. in den Naturwissenschaften — **Leipzig, Germany**
Doctoral student — *2022–2023*
Non-linear algebra group. Withdrew for personal reasons.

Dept. of Mathematics, The University of Auckland — **Auckland, NZ**
Research assistant — *2020–2021*
- Construction and symmetry properties of spherical (t, t)-designs (funded by Dr. Shayne Waldron)
- Teaching of semester-long graduate seminar on Kleinian groups (funded by Dr. Jeroen Schillewaert)

Employment history (teaching)

School of Mathematics, Monash University **Melbourne, Australia**
Sessional teaching associate *2025–2026*

- Tutoring and marking MAT 1030 (service calculus), sem. 1, 2026
- Tutoring and marking MAT 1020 (service calculus), sem. 1, 2025

Dept. of Mathematics, The University of Auckland **Auckland, NZ**
Graduate teaching assistant *2019–2022*

- Marking MATHS 208 (multivariate calculus service course), summer sem., 2022
- Tutoring & marking MATHS 332 (real analysis); tutoring MATHS 190 (general education course), sem. 2, 2021
- Tutoring & marking MATHS 250 (calculus & linear algebra II); tutoring COMPSCI 225 (discrete mathematics for CS); marking MATHS 254 (number theory; groups; metric spaces), sem. 1, 2021
- Tutoring & marking MATHS 190 (general education course); tutoring MATHS 120 (linear algebra); marking MATHS 340 (real & complex calculus), sem. 2, 2020
- Tutoring MATHS 162 (computational mathematics & combinatorics); assistance room tutor; marking MATHS 120 (linear algebra), MATHS 130 (calculus), COMPSCI 120 (mathematics for CS), sem. 1, 2020
- Marking MATHS 253 (calculus & linear algebra III), sem. 2, 2019

MyTuition **Auckland, NZ**
Secondary school tutor *2017–2019*
Mathematics, physics, chemistry, geography

Awards and scholarships

Program associate fellowship, SLMath: Topological and Geometric Structures in Low Dimensions Program (Jan. to Mar. 2026)

CARMA-MATRIX Maths Art Poster Competition: first prize in the outreach category for *Fractals that parameterise hyperbolic handlebodies* (2025)

Research Training Program Scholarship and Stipend (Australian Govt.): tenure 2024–2027

Clay Mathematics Institute Early Career Researcher Support: for attendance of NZMRI Summer School on Groups and Dynamics, Nelson (2023)

Kalman Summer Scholarship: for attendance of NZMRI summer meeting on number theory, Akaroa (2022)

Uni. of Auckland Dept. of Mathematics Student Research Conference prize: (2021)

Kalman Summer Scholarship: for attendance of NZMRI summer meeting, Napier (2021)

Uni. of Auckland Postgraduate Honours Scholarship: tenure 2020

Uni. of Auckland Summer Research Scholarship: for *Numerical construction of spherical* (t, t)*-designs* with Dr. Shayne Waldron (2019)

Uni. of Auckland Faculty of Arts Dean's List: (2017)

NZQA Outstanding Scholar Award: Limited to top 40–60 secondary school students in New Zealand, including an Outstanding Scholarship in Classical Studies (one of 13 awarded nationally) (2016)

Royal Society of New Zealand scholarship: to attend XVI Summer Research School in Mathematics and Informatics, Blagoevgrad, Bulgaria (2016)

Academic works

Preprints and submitted

[1] A. Elzenaar, Expansion joints in hyperbolic manifolds, 2025. arXiv 2512.00879.

[2] A. Elzenaar, From disc patterns in the plane to character varieties of knot groups, 2025. arXiv 2503.13829.

[3] A. Elzenaar, Peripheral subgroups of Kleinian groups, 2025. arXiv 2508.00297.

[4] A. Elzenaar, Changing topological type of compression bodies through cone manifolds, 2024. arXiv 2411.17940.

[5] A. Elzenaar, J. Gong, G. J. Martin, and J. Schillewaert, Bounding deformation spaces of Kleinian groups with two generators, 2024, to appear in *Annales de l'Institut Fourier*. arXiv 2405.15970.

[6] A. Elzenaar, G. J. Martin, and J. Schillewaert, On thin Heckoid and generalised triangle groups in $\mathrm{PSL}(2,\mathbb{C})$, 2024. arXiv 2409.04438.

Published

[7] A. Elzenaar, G. J. Martin, and J. Schillewaert, The combinatorics of Farey words and their traces, *Groups, Geometry, and Dynamics* **20** no. 1 (2026), 215–256. MR 5032293. Zbl 8166380. arXiv 2204.08076. `https://doi.org/10.4171/GGD/832`.

[8] A. Elzenaar and S. Waldron, Putatively optimal projective spherical designs with little apparent symmetry, *Journal of Combinatorial Designs* **33** no. 6 (2025), 222–234. MR 4895104. Zbl 1562.05025. arXiv 2405.19353. `https://doi.org/10.1002/jcd.21979`.

[9] A. Elzenaar, G. J. Martin, and J. Schillewaert, Concrete one complex dimensional moduli spaces of hyperbolic manifolds and orbifolds, in *2021-22 MATRIX annals* (D. R. Wood, J. de Gier, and C. E. Praeger, eds.), Springer, 2024, pp. 31–74. MR 4807267. Zbl 1520.30060. arXiv 2204.11422.

[10] A. Elzenaar, G. J. Martin, and J. Schillewaert, Approximations of the Riley slice, *Expositiones Mathematicae* **41** no. 1 (2023), 20–54. MR 4557273. Zbl 1520.30060. arXiv 2111.03230. `https://doi.org/10.1016/j.exmath.2022.12.002`.

Selected talks

- "Quantifying shape and position of discreteness loci in $\mathrm{PSL}(2,\mathbb{C})$", Postgraduate seminar, Topological and Geometric Structures in Low Dimensions (SLMath, Berkeley), 13 March 2026.
- "Generalised Ford domains arising from peripheral geometry", 69th Annual Meeting of the Australian Mathematical Society (La Trobe Uni.), 9 December 2025.
- "What do knots have to do with algebra?", 2025 NZMS Colloquium (Uni. of Waikato), 27 November 2025.
- "Discontinuous subgroups of come in real-algebraic families with stable combinatorics", 9th Australian Algebra Conference (La Trobe Uni.), 18 November 2025.
- "Cone manifolds and compression bodies", Topology Seminar (Monash Uni.) 22 October 2025.
- "Deformations of 3-orbifold holonomy groups and applications", Early Career Showcase in Low-Dimensional Topology, Joint Meeting of the NZMS, AustMS, and AMS (Uni. of Auckland), 11 December 2024.
- "Combinatorial structures in trace polynomials of function groups", 8th Australian Algebra Conference (ANU, Canberra), 28 November 2024.
- "Two-bridge knots, genus two surfaces, and discrete groups with two generators", Hodgsonfest: Geometry and topology in low dimensions (Uni. Melbourne) 12 November 2024.
- "Is $\mathrm{PSL}(2,\mathbb{C})$ discrete?", Topology Seminar (Monash Uni.) 24 July 2024.
- "The dynamic in the static: Manifolds, braids, and classical number theory", Regiomontanus PhD Seminar (Uni. Leipzig), 10 May 2023.

- “What is a Kleinian group?”, Australian Postgraduate Algebra Colloquium, 21 September 2022.
- “Pictures of hyperbolic spaces”, Discrete Mathematics and Geometry Seminar (TU Berlin), 4 May 2022.
- “Strange circles: The Riley slice of quasi-Fuchsian space”, Seminar on Nonlinear Algebra (MPI MiS), 27 April 2022.
- “Approximating the Riley slice exterior”, Matrix Inst. workshop on Groups and Geometries, 2 December 2021.
- “Some properties of 2×2 matrices”, Dept. of Mathematics Student Research Conference (Uni. of Auckland), 8 June 2021.
- “Real varieties of spherical designs”, Algebra and Combinatorics Seminar (Uni. of Auckland), 1 April 2021.

Professional service

- Refereeing for scholarly publications.
- Reviewing for zbMATH Open.

Selected events organised

- Minicourse on knot theory and geometry (Uni. Auckland), including guest lectures by Lavender Marshall and Josh Lehman (July 2023).
- Reproducibility in Computer Algebra event (MPI-MiS/TU Berlin), co-organised with Christiane Görgen and Lars Kastner (August 2022).
- A Day of Geometry and Lorentzian Polynomials (MPI-MiS), single day of short informal talks in preparation for Göran Gustafsson Symposium at the Institut Mittag-Leffler (May 2022).
- Seminar on Kleinian groups (Uni. Auckland), weekly lectures covering approximately the first half of *Kleinian Groups* by B. Maskit along with some sections of Thurston’s lecture notes and the book *Hyperbolic Manifolds and Discrete Groups* by M. Kapovich. Funded by Jeroen Schillewaert through a Research Assistantship (2021).
- Reading group on algebraic geometry (Uni. Auckland), jointly organised with Oliver Li (2020–2021).

Professional affiliations

- New Zealand Mathematical Society, Student member
- Australian Mathematical Society, Student member

References

Teaching and research reference letters available on request.

[March 12, 2026]